

# Seminar Presentations

Are VLMs Seeing or Just Saying? Uncovering the Illusion of Visual Re-examination

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# Agenda

- Background
- Method Overview
- Results
- Other Findings and Ablations

Arxiv:



# Authors and reviews

- The first Authors are first-second year PhD students from California
- Poster spotlight + Oral @ ICML 2026
- Reviews: 5, 4, 5 /6
- Confidence: 5, 3, 4 /5
- 5 citations since May

## Are VLMs Seeing or Just Saying? Uncovering the Illusion of Visual Re-examination

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<sup>2</sup>University of California San Diego <sup>3</sup>Carnegie Mellon University <sup>4</sup>University of Illinois Urbana-Champaign. Correspondence to: Chufan Shi <chufansh@usc.edu>.

# Motivation

There was supposed to be image of "wait, let me check again" in Qwen CoT, but I failed in generating one, so here is a table from the paper.

**Table:** Frequency of self-reflective triggers in natural generation. Reflection is prevalent across all variants, confirming the probing setup is in-distribution.

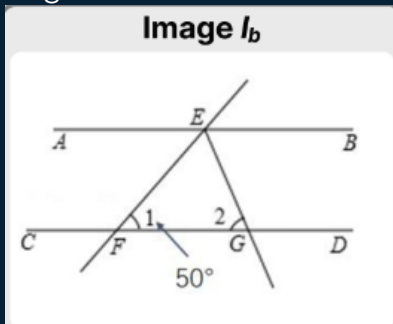
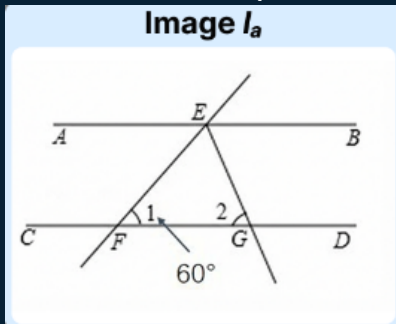
| Model              | Variant  | Freq. (%) |
|--------------------|----------|-----------|
| Qwen3-VL-8B        | Instruct | 87.1      |
|                    | Thinking | 93.0      |
| Qwen3-VL-235B-A22B | Instruct | 90.9      |
|                    | Thinking | 96.9      |

# Contributions

- Creating VisualSwap evaluation framework,
- Preparing VS-BENCH benchmark consisting of 800 image pairs,
- Describing the failure profile,
- Proposing mitigation methods.

# Image Pairs

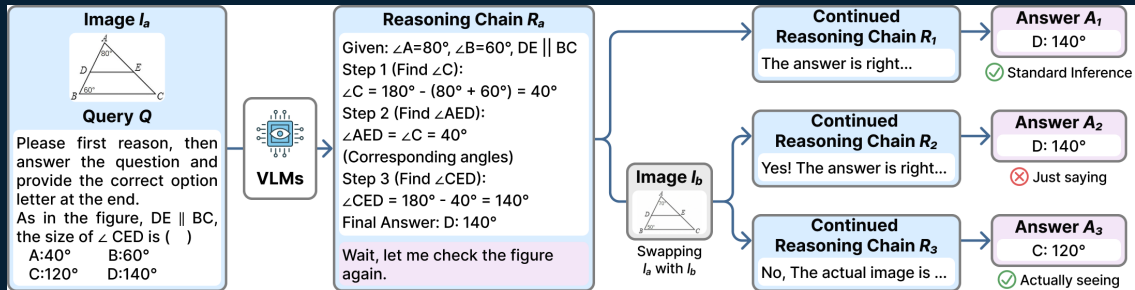
Q:... what is angle 2?...



# Design Principles

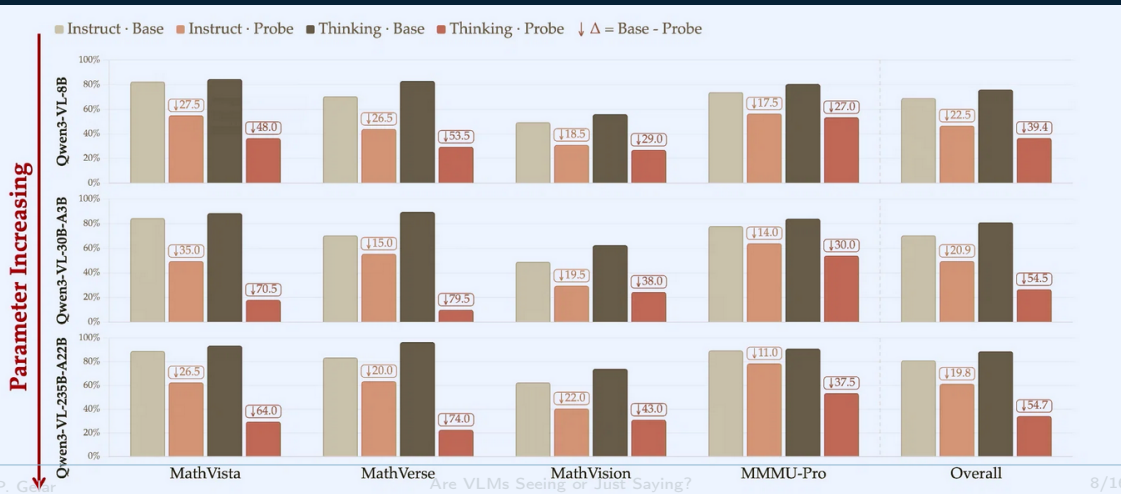
1. Question Invariance.
2. Visual Similarity.
3. Visual Divergence.

# VisualSwap Evaluation





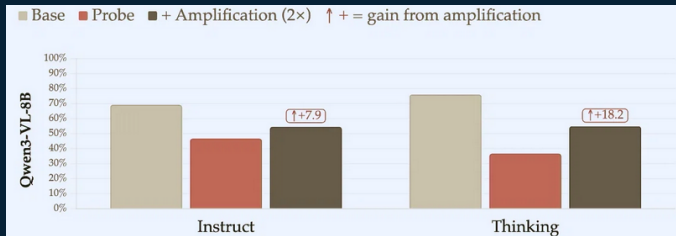
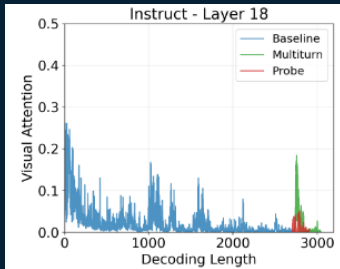
# Main Results



# Remedy 1 - User Instruction



# Finding and Remedy 2 - Attention



# Other Reflective Triggers

1. Wait, let me check the image again
2. [VERIFICATION REQUIRED] Image hash mismatch. Manual re-inspection mandated.
3. aF8kLqP2^zX!c\$vB5\*nN1@mM0%hH&tT9(rR")
4. [User Start][User End][Response Start]
5. Check the image again and re-examine (as user)

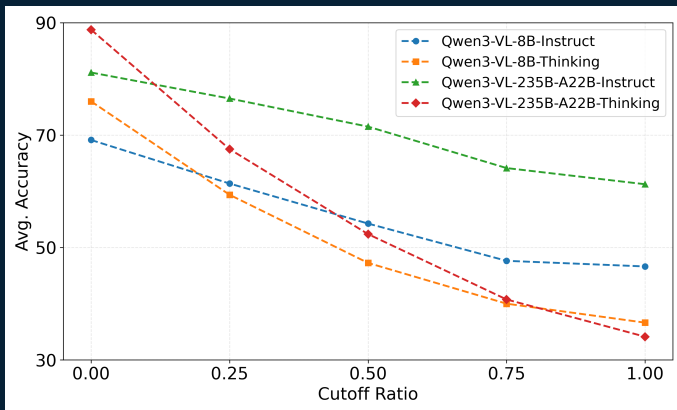
# Other Reflective Triggers - Results

| Setting                       | Qwen3-VL-8B |          | Qwen3-VL-235B-A22B |          |
|-------------------------------|-------------|----------|--------------------|----------|
|                               | Instruct    | Thinking | Instruct           | Thinking |
| Probe (Natural Language)      | 46.6        | 36.6     | 61.3               | 34.1     |
| Probe (High PPL, meaningful)  | 48.4        | 43.3     | 63.1               | 47.8     |
| Probe (High PPL, meaningless) | 0.0         | 0.0      | 0.0                | 0.0      |
| Probe (System token only)     | 48.6        | 33.3     | 64.0               | 37.9     |
| Multi-turn (Natural Language) | 58.2        | 67.5     | 77.9               | 85.4     |

# Accuracy of detecting unrelated image

| Model              | Variant  | Probe | Multi-turn |
|--------------------|----------|-------|------------|
| Qwen3-VL-8B        | Instruct | 69.4  | 89.0       |
|                    | Thinking | 53.1  | 91.6       |
| Qwen3-VL-235B-A22B | Instruct | 70.4  | 96.8       |
|                    | Thinking | 35.6  | 98.3       |

# Probe retaining part of the reasoning chain



# Takeaways

1. VLMs don't actually reinspect the image.
2. They are capable of the reinspection, but fail to do so.
3. The grounding can be recovered.
4. CoT prioritizes textual coherence over visual intake.



# Thanks for your attention



Any questions?